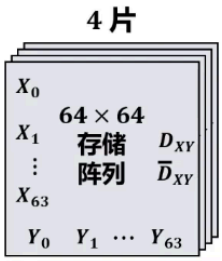
一、存储器结构

1.1 规格介绍

【举例】

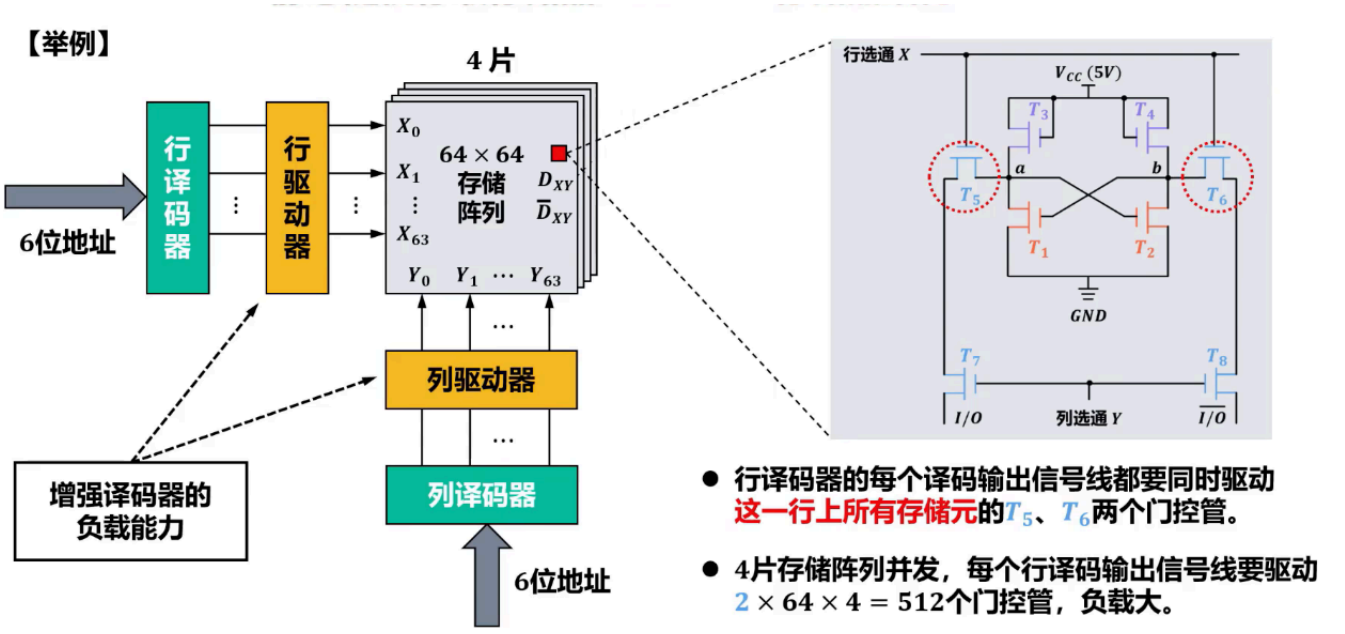


- 每个 64×64 存储阵列（即64行64列）包含$64 \times 64 = 4096$个存储元（每个存储元存储1位二进制数）
- 行选通线：$X_0 \sim X_{63}$，共64条
- 列选通线：$Y_0 \sim Y_{63}$，共64条
- 存储字长：4位（一次可存取的位数）
- 存储容量：4096位/片 $\times$ 4片$= 4096b \times 4 \div 8 = 2$kB

1.2 具体结构

行译码器（可参考[[4.4 主存的结构]]）

【举例】

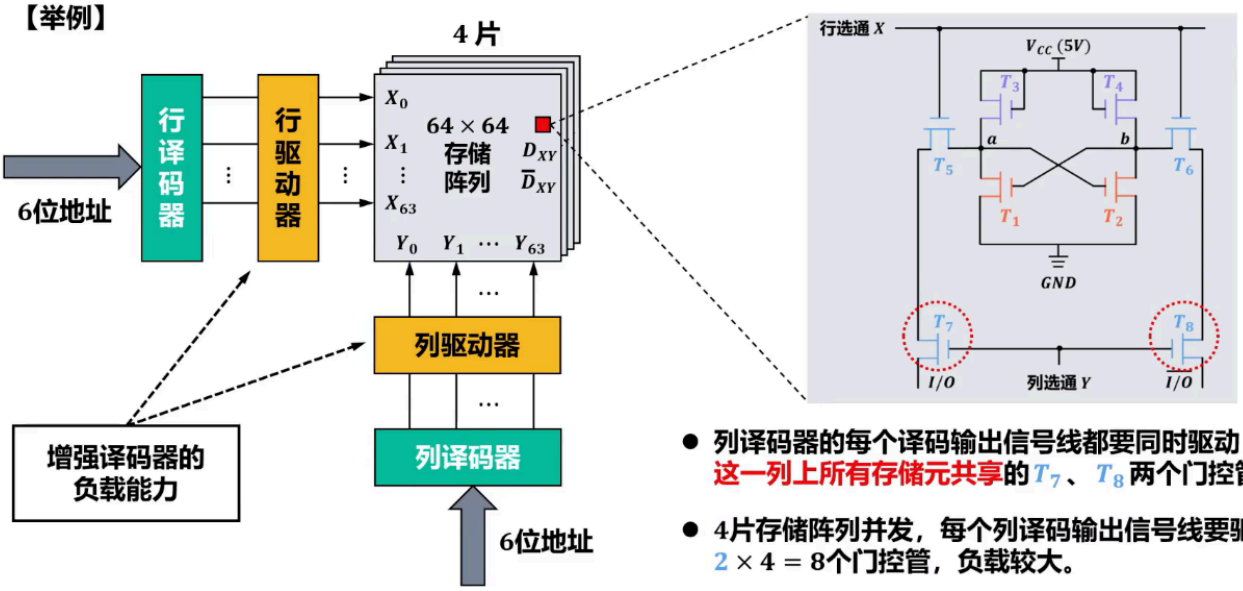


- 行译码器的每个译码输出信号线都要同时驱动这一行上所有存储元的T_5、T_6两个门控管。
- 4片存储阵列并发，每个行译码输出信号线要驱动$2 \times 64 \times 4 = 512$个门控管，负载大。

列译码器

静态随机存取存储器SRAM —— 存储器结构

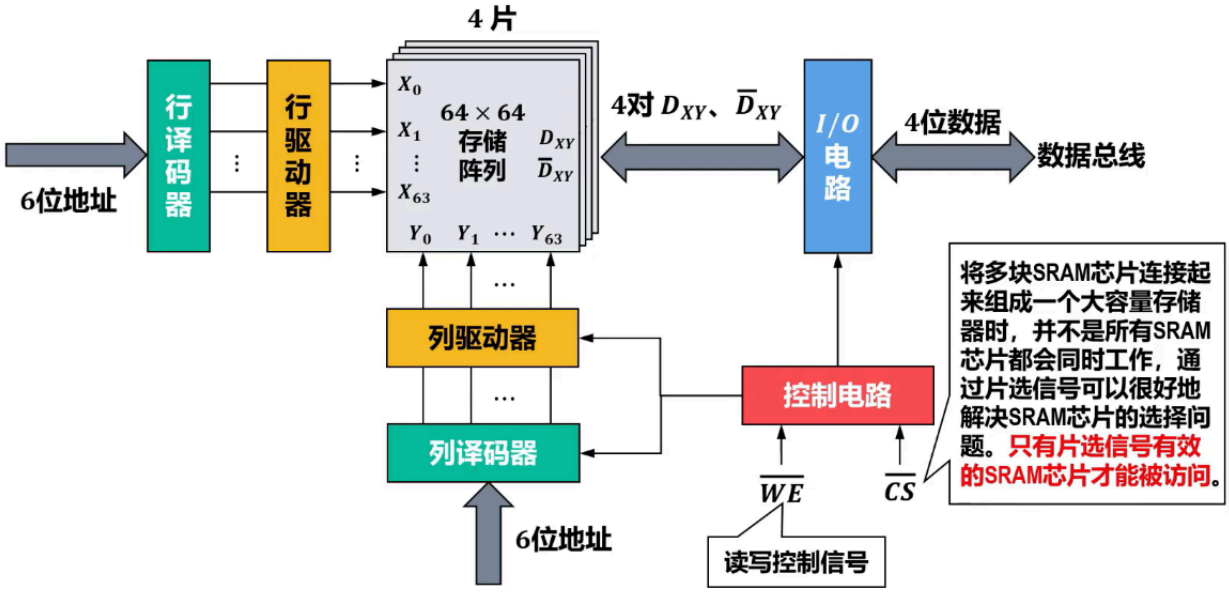
【举例】



- 列译码器的每个译码输出信号线都要同时驱动这一列上所有存储元共享的T_7、T_8两个门控管。
- 4片存储阵列并发，每个列译码输出信号线要驱动$2 \times 4 = 8$个门控管，负载较大。

I/O及控制设备

【举例】



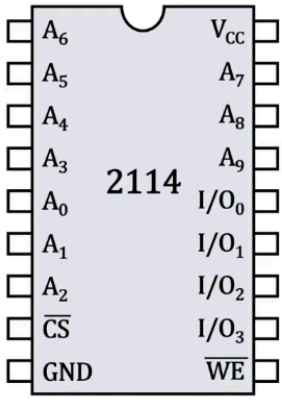
二、芯片实例

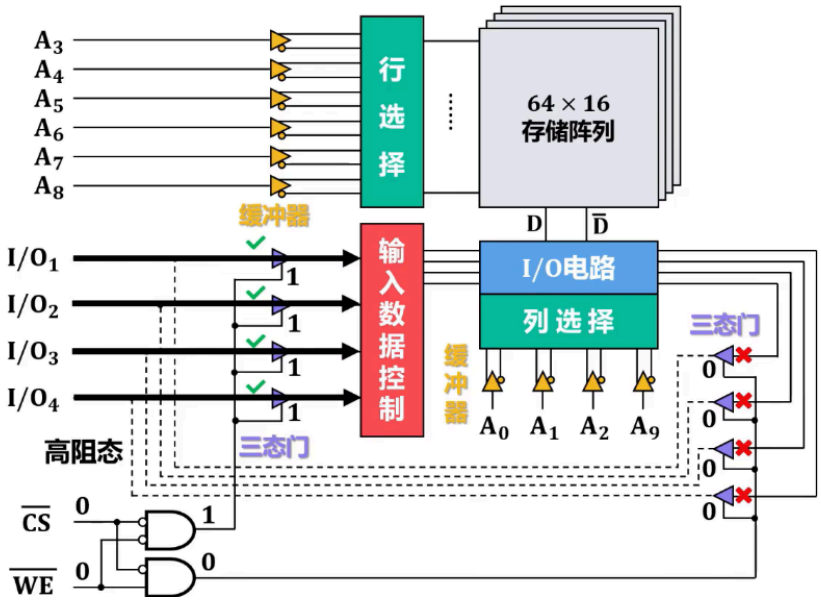
1. 英特尔芯片实例

CS和WE都为有效信号，进行写入操作

静态随机存取存储器SRAM —— 存储器芯片实例

【举例】Intel 2114是英特尔公司推出的一款$1K \times 4$位的SRAM芯片。

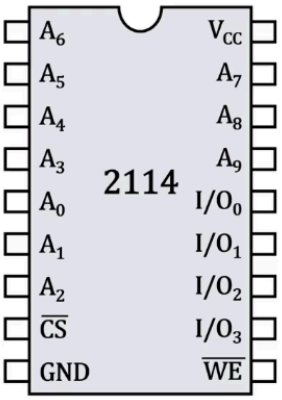


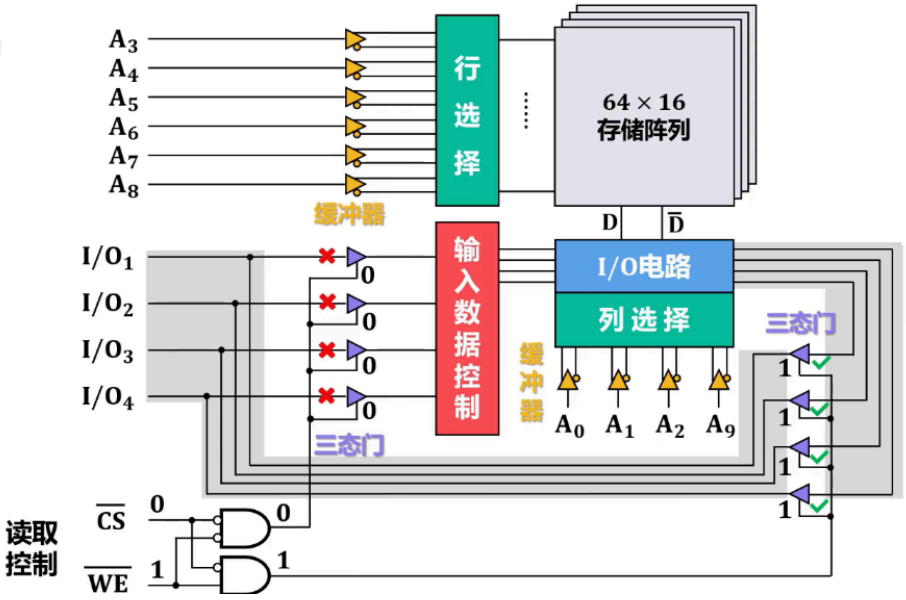


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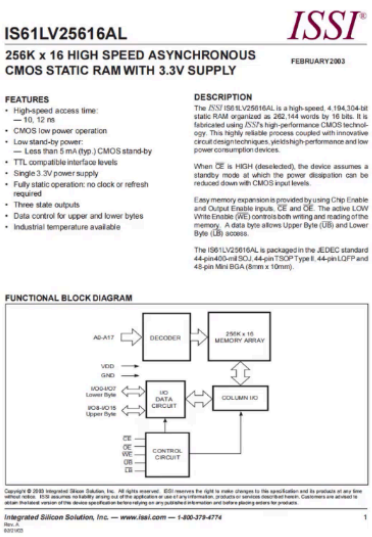




注：WE和CS的有效信号为0，无效信号为1

2. 关于芯片的说明书

功能框架



IS61LV25616AL

ISSI®

256K x 16 HIGH SPEED ASYNCHRONOUS CMOS STATIC RAM WITH 3.3V SUPPLY

FEBRUARY2003

FEATURES

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  — 10, 12 ns
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- Single 3.3V power supply
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- Three state outputs
- Data control for upper and lower bytes
- Industrial temperature available

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FUNCTIONAL BLOCK DIAGRAM

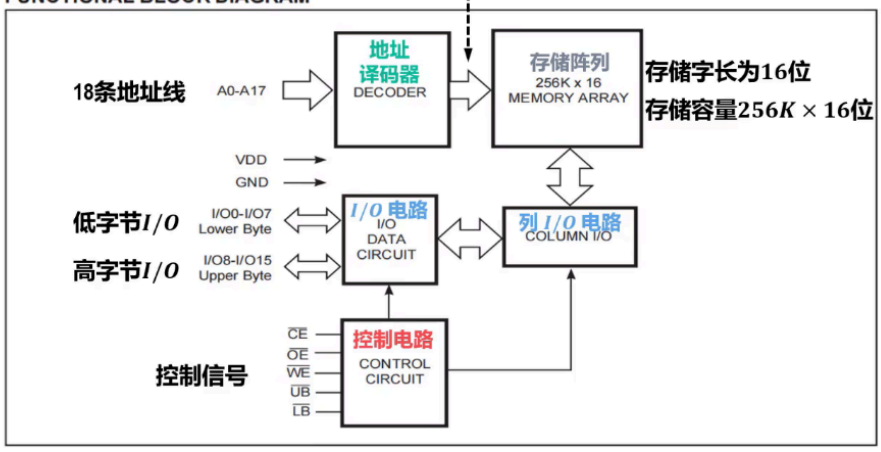
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Data sheet

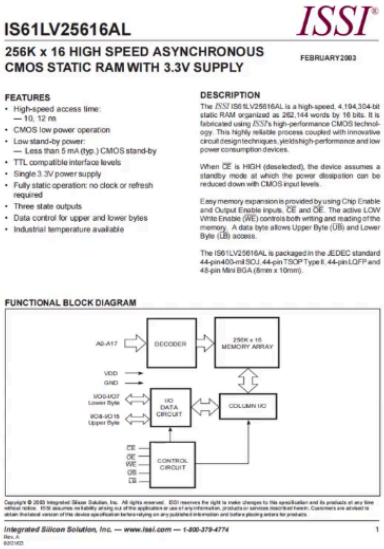
功能框图

$2^{18} = 256K$条译码输出线

FUNCTIONAL BLOCK DIAGRAM



真值表



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Data sheet

真值表 TRUTH TABLE

写使能	芯片使能	输出使能	低位字节控制	高位字节控制

High-Z：高阻态

D_{OUT}：数据输出

D_{IN}：数据输出

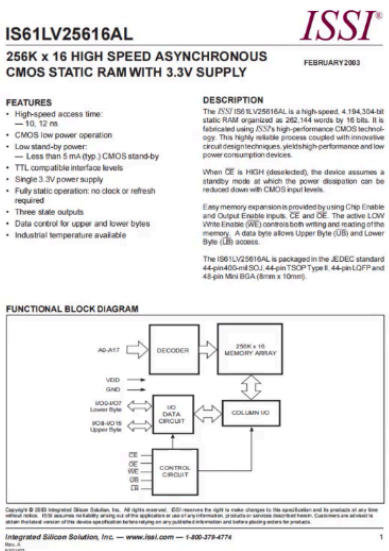
模式 Mode	控制引脚 $\overline{WE}$	$\overline{CE}$	$\overline{OE}$	$\overline{LB}$	$\overline{UB}$	I/O PIN I/O0-I/O7	I/O引脚 I/O8-I/O15	V_{DD} Current
Not Selected 未选中	X	H	X	X	X	High-Z	High-Z	I_{SB1}, I_{SB2}
Output Disabled 输出禁止	H	L	H	X	X	High-Z	High-Z	I_{CC}
	X	L	X	H	H	High-Z	High-Z	
Read 读操作	H	L	L	L	H	D_{OUT}	High-Z	I_{CC}
	H	L	L	H	L	High-Z	D_{OUT}	
	H	L	L	L	L	D_{OUT}	D_{OUT}	
Write 写操作	L	L	X	L	H	D_{IN}	High-Z	I_{CC}
	L	L	X	H	L	High-Z	D_{IN}	
	L	L	X	L	L	D_{IN}	D_{IN}	

X：高电平或低电平都无影响

H：高电平

L：低电平

读写周期



IS61LV25616AL

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每个型号的SRAM芯片有其特定的读写周期特性，只有按照其读写周期去访问SRAM芯片，才能保证读写操作的正确性。

静态随机存取存储器SRAM —— 存储器芯片实例

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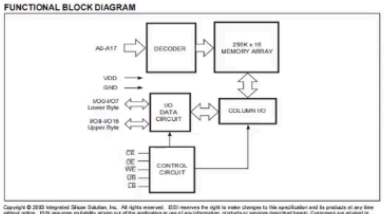
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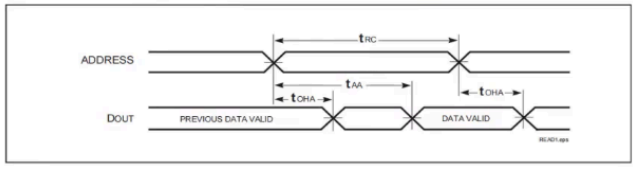


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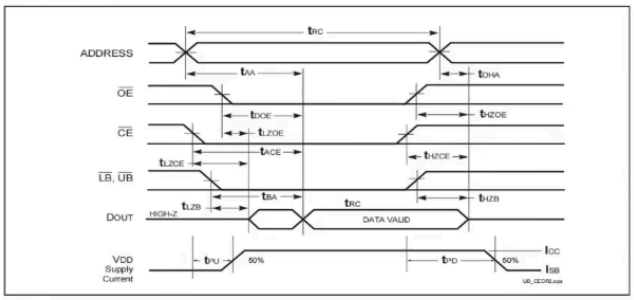
读周期 （模式 1 ）

READ CYCLE NO. 1[1,2] (Address Controlled) ($\overline{CE}$ = $\overline{OE}$ = V_{IL}, $\overline{UB}$ or $\overline{LB}$ = V_{IL})



读周期 （模式 2 ）

READ CYCLE NO. 2[1,3]



Notes:
1. $\overline{WE}$ is HIGH for a Read Cycle.
2. The device is continuously selected. $\overline{OE}$, $\overline{CE}$, $\overline{UB}$, or $\overline{LB}$ = V_{IL}.
3. Address is valid prior to or coincident with $\overline{CE}$ LOW transition.

电源特性

静态随机存取存储器SRAM —— 存储器芯片实例

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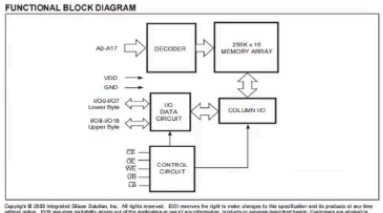
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芯片封装

工作范围

OPERATING RANGE

Range	Ambient Temperature	V_{DD} 10ns	V_{DD} 12ns
Commercial	0°C to +70°C	3.3V +10%, -5%	3.3V ± 10%
Industrial	–40°C to +85°C	3.3V +10%, -5%	3.3V ± 10%

直流电气特性

DC ELECTRICAL CHARACTERISTICS (Over Operating Range)

Symbol	Parameter	Test Conditions		Min.	Max.	Unit
V_{OH}	Output HIGH Voltage	V_{DD} = Min., I_{OH} = –4.0 mA		2.4	—	V
V_{OL}	Output LOW Voltage	V_{DD} = Min., I_{OL} = 8.0 mA		—	0.4	V
V_{IH}	Input HIGH Voltage			2.0	V_{DD} + 0.3	V
V_{IL}	Input LOW Voltage[1]			–0.3	0.8	V
I_{LI}	Input Leakage	GND ≤ V_{IN} ≤ V_{DD}	Com. Ind.	–2 –5	2 5	µA
I_{LO}	Output Leakage	GND ≤ V_{OUT} ≤ V_{DD} Outputs Disabled	Com. Ind.	–2 –5	2 5	µA

Notes:
1. V_{IL} (min.) = –2.0V for pulse width less than 10 ns.

静态随机存取存储器SRAM —— 存储器芯片实例

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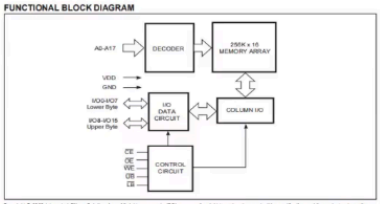
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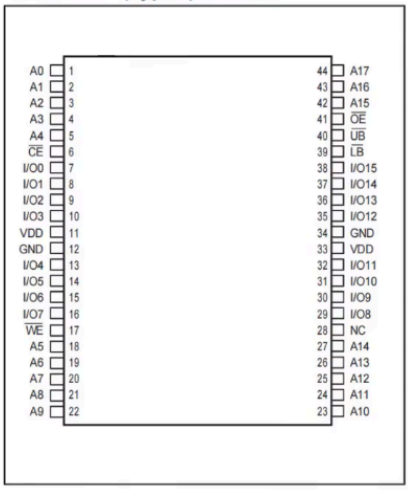
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Data sheet

引脚配置

PIN CONFIGURATIONS

44-Pin TSOP (Type II) and SOJ



PIN DESCRIPTIONS

A0-A17	Address Inputs
I/O0-I/O15	Data Inputs/Outputs
$\overline{CE}$	Chip Enable Input
$\overline{OE}$	Output Enable Input
$\overline{WE}$	Write Enable Input
$\overline{LB}$	Lower-byte Control (I/O0-I/O7)
$\overline{UB}$	Upper-byte Control (I/O8-I/O15)
NC	No Connection
V_{DD}	Power
GND	Ground